De minimis assessments
Supporting document `IndigoCarbon_US-1_2024_0010`

April 29, 2025

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1 Methane (CH_4) Emissions from Upland Crop Production

For the Project, we assessed the potential de minimis standing of methane (CH_4) emissions from upland crop production.

- Methane (CH_4) emissions from upland crop production were determined to be de minimis, and this sources were not included in project carbon credit calculations. This approach to CH_4 upland emissions (literature and expert testimonials) was approved by the Climate Action Reserve (CAR) on October 5th, 2021. Our de minimis assessment assumed all upland crop production soils were non-wetland, and wetland soils were excluded from the Project. Thus, all Project soils are assumed to be upland.

1.1 The proposal to car:

"We have conducted research into the CH_4 emissions from upland soils and are looking to seek approval for a de-minimis demonstration attached. It includes:"

1. We will account for all project CH_4 emissions from livestock and biomass burning (using the protocol default equations, not models).
2. The biogeochemical models are not yet able to be validated for soil CH_4 dynamics on cropland. However, we believe the emissions from soil CH_4 to be de-minimis, because: - Non-wetland soils are generally a net CH_4 sink; and, - Regenerative agricultural practices tend to strengthen, rather than weaken this sink.
3. Thus, we should be allowed to use biogeochemical models that are not validated for soil CH_4. Note; Indigo will be seeking soil organic carbon (SOC) credits on such lands. A review of the literature and expert testimonial supported the position that CH_4 emissions from upland cropland is de-minimis. Continuation of upland crop production and farm management itself does not contribute meaningfully to methane emissions, and research indicates that it may be a net sink. It can be noted that land conversion from un-managed forest to agricultural land may reduce the methane sink."

1.2 CAR's response:

- *"The Reserve will allow this with the understanding that the de-minimis assessment will be detailed in the documentation submitted in verification, and that the assessment will have received expert review and approval*

that will also be detailed in the documentation submitted in verification, such that the verification body will have a chance to review and confirm the de-minimis requirements have been met."

Details from this review are provided in the tables below.

- Table 1: SME Testimonials - documents subject matter expert (SME) commentary supporting the position that methane emissions from upland cropland production is de-minimis. Direct quotes are provided from email exchanges with the relevant SMEs, along with supporting citations. We gathered testimonials from three SMEs for this process.
- Table 2: Additional citations - provides literature sources that support the position that methane emissions from upland cropland production is de-minimis and/or documentation that agricultural soils can serve as a net sink.

Table 1: SME Testimonials

Name/Affiliation/ Contact Info.	**Testimonial**	**Suggested Citations**
Dr. Ankur Desai Professor, University of Wisconsin - Madison Dept of Atmospheric and Oceanic Sciences https://flux.aos.wisc.edu desai@aos.wisc.edu	Generally we've focused on places where there is likely to be high signal - which for CH4, would involve wet/anoxic locations. Upland soils can be methane sources or sinks, depending on moisture conditions and bacteria populations. I am not aware of specific literature on upland crop CH4 emissions, though we have taken some in a few of our fields. There have been a few on forest upland soils, which tend to be near neutral to small sinks (order of magnitude smaller than wetland emissions), so your assumption of de minimus in an unmanaged dry soil system is reasonable. However, application of manure and fertilizer and irrigation would likely change all that.	Yu, X., Millet, D. B., Wells, K. C., Henze, D. K., Cao, H., Griffis, T. J., Kort, E. A., Plant, G., Deventer, M. J., Kolka, R. K., Roman, D. T., Davis, K. J., Desai, A. R., Baier, B. C., McKain, K., Czarnetzki, A. C., and Bloom, A. A.: Aircraft based inversions quantify the importance of wetlands and livestock for Upper Midwest methane emissions, Atmos. Chem. Phys. Discuss., https://doi.org/10.5194/acp2020-826, in review, 2020
Dr. Jonathan Sanderman Associate Scientist, Woodwell Climate Research Center 149 Woods Hole Road, Falmouth, MA 02540 USA jsanderman@woodwellclimate.org	My general understanding is upland soils are small net sinks of methane and if anything degraded systems are smaller sinks so regenerative practices should enhance the methane uptake (this was shown from a broader land use change perspective in a recent metaanalysis (see citation). If you generally buy that argument which I think there is strong theoretical support for (greater aeration, more active microbial communities), then I would consider methane emissions to certainly be de minimus.	Wu, J, Chen, Q, Jia, W, et al. Asymmetric response of soil methane uptake rate to land degradation and restoration: Data synthesis. Glob Change Biol. 2020; 26: 6581– 6593. https://doi.org/10.1111/gcb.15315

Table 1: SME Testimonials

Name/Affiliation/ Contact Info.	**Testimonial**	**Suggested Citations**
Eric Toensmeier, toensmeier@gmail. com http://www.perenn ialsolutions.org/me et-eric-toensmeier-perennial-solutions-edible-permaculture-books-videos-workshops-organic-gardening.html	According the FAO Statistical Service, at the global level methane emissions from burning crop residues are 22 MMT CO2-eq/yr (quite low compared to say methane from rice production, which is 533 MMT/yr). No other CH4 emissions from upland crop production that I know of. In the interest of completion I should mention that there are also rather modest methane sinks in soils. When land is converted from forest or grassland to cropland, 2/3 of this sink capacity is lost. It takes a century or so to be rebuilt. Use of compost and increased retention of crop residues can also make a small gain to this already quite modest sink.	Smith, K. A., et. al. (2000). Oxidation of atmospheric methane in Northern European soils, comparison with other ecosystems, and uncertainties in the global terrestrial sink. Global Change Biology, 6(7), 791- 803. Singh, J. S., & Gupta, V. K. (2016). Degraded land restoration in reinstating CH4 sink. Frontiers in microbiology, 7, 923

Table 2: Additional Citations

Name/Affiliation/Contact Info.
Yu, X., Millet, D. B., Wells, K. C., Henze, D. K., Cao, H., Griffis, T. J., Kort, E. A., Plant, G., Deventer, M. J., Kolka, R. K., Roman, D. T., Davis, K. J., Desai, A. R., Baier, B. C., McKain, K., Czarnetzki, A. C., and Bloom, A. A.: Aircraft-based inversions quantify the importance of wetlands and livestock for Upper Midwest methane emissions, Atmos. Chem. Phys. Discuss., https://doi.org/10.5194/acp-2020-826, in review, 2020.
Wu, J, Chen, Q, Jia, W, et al. Asymmetric response of soil methane uptake rate to land degradation and restoration: Data synthesis. Glob Change Biol. 2020; 26: 6581– 6593. https://doi.org/10.1111/gcb.15315
Smith, K. A., et. al. (2000). Oxidation of atmospheric methane in Northern European soils, comparison with other ecosystems, and uncertainties in the global terrestrial sink. Global Change Biology, 6(7), 791-803.
Singh, J. S., & Gupta, V. K. (2016). Degraded land restoration in reinstating CH4 sink. Frontiers in microbiology, 7, 923
Ho, A., Reim, A., Kim, S.Y., Meima-Franke, M., Termorshuizen, A., de Boer, W., van der Putten, W.H. and Bodelier, P.L.E. (2015), Unexpected stimulation of soil methane uptake as emergent property of agricultural soils following bio-based residue application. Glob Change Biol, 21: 3864-3879. https://doi.org/10.1111/gcb.12974
K.E. Dobbie, K.A. Smith, A. Prieme´, S. Christensen, A. Degorska, P. Orlanski, Effect of land use on the rate of methane uptake by surface soils in Northern Europe, Atmospheric Environment, Volume 30, Issue 7, 1996, Pages 1005-1011, ISSN 1352-2310, https://doi.org/10.1016/1352-2310(95)00416-5.

Table 2: Additional Citations

Name/Affiliation/Contact Info.
Carlson, K., Gerber, J., Mueller, N. et al. Greenhouse gas emissions intensity of global croplands. Nature Clim Change 7, 63–68 (2017). https://doi.org/10.1038/nclimate3158
Reay, D., Smith, P., Christensen T.R., James R.H., Clark H. Methane and Global Environmental Change. Annu. Rev. Environ. Resour. 2018. 43:165–92. https://doi.org/10.1146/annurev-environ-102017- 030154
Drawdown.org: Farming our way out of the climate crisis.

2 Summary of Changes

Section	**Date**	**Summary of Changes**
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